Project Presentation of Co-Op Project at Industry (Module-1)

On

ARENA

A Minimalist Survival Arcade Game Based on Skill and Reflex

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Introduction & Objective

ARENA is a minimalist survival arcade game focused on reflex, precision, and fast-paced gameplay. Players control a sphere and survive against increasing enemy pressure. The project emphasizes skill-based challenge, responsive controls, and optimized mobile performance.

Objective

- **Develop a skill-based survival game.**
- Create engaging gameplay with increasing difficulty.
- Ensure smooth, low-CPU performance.
- Provide responsive touch controls.
- Deliver a lightweight and scalable mobile game.

Methodology

- **Designed the game using a modular development approach.**
- **Implemented core mechanics including player movement, enemy spawning, and collision detection.**
- **Used iterative development for testing gameplay balance and responsiveness.**
- **Applied performance optimization techniques for smooth execution.**
- **Evaluated the system through functionality and gameplay testing.**

Key Findings

- Minimalist mechanics can create engaging gameplay.
- Optimized architecture improves performance and responsiveness.
- Skill-based difficulty increases replayability.
- Lightweight design supports smooth gameplay on mobile devices.
- Simple mechanics combined with strong visuals enhance user experience.

Conclusion

ARENA demonstrates that a minimalist, skill-based arcade game can deliver engaging gameplay through optimized performance, responsive controls, and escalating challenge. The project highlights how simple mechanics and efficient engineering can create a scalable and compelling mobile gaming experience.

Future Scope

- **Add multiplayer and leaderboard features.**
- **Introduce adaptive enemy AI and new game modes.**
- **Expand with power-ups and procedural challenges.**
- **Optimize further for cross-platform deployment.**
- **Scale the project into a full commercial game product.**

References

- React Native Official Documentation
- Expo SDK Documentation
- TypeScript Handbook
- Game Programming Patterns — Robert Nystrom
- Mobile Game Development and Performance Optimization Research Papers

Thank You